

Project Y

Los Alamos • 1943–45

15 STAGES • 64 STOPS • GRADE 12 • PROJECTY

THE OPENING CARD

It is April 1943, and the site is three weeks old: a boys' school on a mesa in New Mexico, with a laboratory going up around it. There is a fence, and most of the people arriving cannot tell their families where they are. You are a scientist newly assigned to Project Y. It has one problem to solve and a schedule set by a war. Five divisions are working on it at once — theory, experiment, chemistry and metallurgy, ordnance, explosives — and not one of them can finish without the other four. Nobody yet knows whether the thing being described on three different blackboards will work at all. Oppenheimer has put a meeting in the calendar every week where anyone may be asked anything, so that being wrong costs days instead of months.

Read the atomic world

BEFORE THE STAGE — GREET

The mesa is full of people who cannot tell you what they do

Everybody here has arrived in the last year and half of them are not allowed to say what they are working on. Oppenheimer, the laboratory director, takes new staff round the divisions once, because compartmentalisation means the only way you will know who to ask is having been introduced to them by somebody who knows what you are cleared for.

PLAN CARD 104W

April 1943, and the site is 3 weeks old. Five divisions, one mesa, and three different ways of writing the same nucleus on three different blackboards. The laboratory director, J. Robert Oppenheimer, has scheduled a colloquium every week for exactly this reason: nothing here can be coordinated until everybody describes matter the same way. Today you settle three things. What a nuclide symbol actually specifies. Why two isotopes of an element act alike in a beaker and differently in a counter. And how radiation becomes a number an instrument can report. Every measurement made here for the next two years rests on these three.

BRIEFING 35W

Project Y cannot coordinate five divisions until everyone describes matter in the same way. Begin with the nucleus, connect isotope notation to chemical identity, and finish by seeing how radiation becomes a measurable electrical signal.

WHAT YOU DECIDE 16W

Build a shared language for nuclei, isotopes, and detector signals before any team begins advanced work.

1.1 The nucleus as a physical system

THEORY & CALCULATIONS · A ROOM · PROTOCOL

SCENE 33W

Your first morning on the Hill. The Theoretical Division hands you four nuclide labels with no explanation and expects you to pull the right quantity out of each one before anybody calculates anything.

GUIDE 52W

Four things you need and four ways to get them. Pair them by asking what each symbol counts. One counts protons. One counts protons and neutrons together. So one of these answers is a subtraction rather than a symbol on the label. And a neutral atom's electron count follows from its protons.

ASK 18W

Match each information need or situation to the best action, instrument, or control. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an atom has a nucleus of protons and neutrons, with electrons around it

VERDICT 91W

Notation is only useful when each symbol is tied to a physical count. The atomic number Z is the proton count, and it fixes which element this is. The mass number A counts protons and neutrons together, so the neutron count is A minus Z — a subtraction, not a symbol on the label. Two atoms are isotopes of one element when Z is the same and A differs. And a neutral atom has as many electrons as protons, which is why Z decides the chemistry as well as the identity.

TAKEAWAY 11W

Nuclear reasoning starts by translating notation into particles and conserved quantities.

1.2 Isotopes and chemical identity

CHEMISTRY & METALLURGY · A PERSON · PROTOCOL

SCENE 33W

Chemistry and Metallurgy is being asked what it can and cannot deliver, and the answer turns on one fact about isotopes. Everything shipped here has both a chemical form and an isotopic composition.

GUIDE 48W

Four questions and four answers, and three of them say yes. Ask of each whether the property lives in the electrons or in the nucleus. Bonding is electrons, and extra neutrons barely disturb them. Stability is nuclear. So ordinary chemistry has almost no handle here, and mass does.

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TAKES AS READ

chemical bonding is done by electrons

VERDICT 84W

Isotopes of an element carry the same proton count and nearly the same electrons, so they form the same compounds and answer to the same reagents. Ordinary chemistry therefore has almost no handle for separating them. Nuclear stability is different: it is set inside the nucleus, and half-lives can differ enormously between isotopes of one element. Mass spectrometry sorts by mass-to-charge ratio, responding to the difference chemistry mostly ignores. That is why isotopic separation is a physical problem as well as a chemical one.

TAKEAWAY 10W

Materials work must track both chemical form and isotopic composition.

1.3 Ionization and detector signals

EXPERIMENTAL PHYSICS & DIAGNOSTICS • EXPERIMENTAL PHYSICS • A ROOM • SEQUENCE

SCENE 36W

The Physics Division cannot see a single neutron or alpha particle, and neither can anybody else. Everything the Hill will claim about cross sections, backgrounds and yields rests on a chain of conversions working in order.

GUIDE 51W

These four are one chain, and a detector never measures the radiation itself. Ask of each card what has to exist before it. Nothing is amplified before something is set free. And nothing is set free before energy is deposited. That middle step decides what the instrument can see at all.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

radiation deposits energy in whatever it passes through • decay constant, half-life and activity — taken as read

VERDICT 86W

A detector measures the consequences of an interaction, never the radiation itself. Energy has to be deposited in the sensitive material first — nothing downstream exists until that happens. That deposit sets free charge or light, which is the transducer step and the one that decides what the instrument is sensitive to at all. Electronics then collect and amplify a response too small to handle raw. The data system digitises and stores a pulse last. Get the chain wrong and you are measuring your own apparatus.

TAKEAWAY 11W

A detector measures the consequences of an interaction, not radiation directly.

A nucleus is defined by its proton and neutron counts; experiments turn nuclear events into signals that can be counted.

Account for nuclear energy

BEFORE THE STAGE — TRIAL

Walk the mesa before you are trusted with an errand on it

The site is a mesa with one road, and the buildings that matter are not the ones with the signs. Walk it once now — theoretical, the physics building, chemistry and metallurgy, the shops — because an errand across this site in the dark with a box you may not open is not the time to be learning it.

PLAN CARD 101W

May 1943. Two architectures are on the table, the site can properly staff one of them, and Groves wants to know which by the end of the month. Today you work out where nuclear energy comes from, check that the instruments reporting it are calibrated, and compare the two paths. Every energy figure quoted on this mesa is really the size of an electrical pulse, converted by a calibration somebody had to establish with a known source. An uncalibrated number is a voltage with ambitions. Choose on preference instead of evidence and a year of work goes into the wrong programme.

BRIEFING 33W

The next step is to understand why nuclear changes involve such large energy scales, how a measurement system assigns an energy value, and why Los Alamos initially pursued more than one technical path.

WHAT YOU DECIDE 11W

Connect mass, binding energy, calibration, and the program's competing early architectures.

2.1 Mass defect and binding energy

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 37W

A theorist at the blackboard wants a number before lunch: how tightly a medium-heavy nucleus is bound. The arithmetic is small enough for one line, but the energy scale is the reason this laboratory exists at all.

GUIDE 48W

Four numbers, and two of them are the same defect written a thousand times apart. Ask of each whether it is per nucleon or per nucleus. The conversion turns mass into energy, so it belongs beside the mass term. And work out the per-nucleon figure before the total.

ASK 11W

Estimate the total binding energy and the binding energy per nucleon.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

mass and energy are related by $E = mc^2$, with E the energy, m the mass and c the speed of light

VERDICT 88W

A bound nucleus weighs less than the loose protons and neutrons that went into it. That missing mass is the binding energy holding it together. The conversion factor is the speed of light squared. So a mass difference far too small to weigh becomes an energy larger than any chemical reaction can produce. Work in atomic mass units and 1 unit is about 931 MeV. That is what turns a defect of 8 thousandths of a unit per nucleon into more than 1000 MeV for the whole nucleus.

TAKEAWAY 9W

Tiny changes in mass correspond to large nuclear energies.

2.2 One reference line is not a calibration

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · DEGENERACY

SCENE 43W

Two Physics groups bring spectra from different instruments, and the peaks refuse to line up. Each group trusts its own axis. An afternoon comparison meeting is approaching, and somebody has to decide whether the disagreement belongs to the samples or to the instruments.

GUIDE 49W

Two controls, gain and offset, slide along a family of calibrations that all put one reference line where it belongs. Find out how wide that family is before deciding anything. Then bring in the second reference line, which does not permit the same trade, and watch the family collapse.

ASK 10W

Which gain and offset are fixed by both reference lines?

BEHIND THE BUTTON 151W

Why one line cannot calibrate an axis. A linear axis has two unknowns: how much a step is worth, and where zero sits. One known peak gives one equation. Every gain paired with the offset that puts that peak in the right place fits equally well, which is exactly the family the controls slide along.

Why the second line is not a repeat. It constrains the same two unknowns through a different combination, so the pairs that satisfied the first no longer satisfy this one. Where the two constraints cross is the calibration, and it is why standards come with more than one line.

What is at stake this afternoon. If the axes can be reconciled, the spectra agree and the disagreement was instrumental. If they cannot, the difference is in the samples. Both groups trusting their own axis is how a meeting spends an afternoon arguing about the wrong thing.

TAKES AS READ

a straight-line calibration can have both a slope and an offset · one equation cannot determine two unknowns

VERDICT 70W

A straight-line calibration has two unknowns: gain and offset. One known line gives only one equation, so many gain-and-offset pairs reproduce it exactly. That is not noise; it is missing information. A second line at a different energy supplies a second equation. The two allowed loci cross at one point, fixing both parameters. This is why a convincing-looking one-point calibration can still place every unknown peak at the wrong energy.

TAKEAWAY 18W

One calibration point cannot determine both a slope and an offset; a second independent point breaks that degeneracy.

2.3 Two weapon architectures at Los Alamos

ORDNANCE & ENGINEERING · A ROOM · PROTOCOL

SCENE 41W

Oppenheimer has asked Ordnance for a recommendation before the next program meeting. Two assembly concepts are still alive, but choosing between them commits people, buildings, and months of work. The room is waiting for a decision that will reshape the Hill.

GUIDE 49W

Four situations and four responses, and both designs chase the same goal. Ask of each whether it is about the geometry of the assembly or about its timing. One route drives two pieces along a line. One squeezes inward, and nobody has built one. A measurement decides between them.

ASK 16W

Match each situation to the best action, control, document, or interpretation. Each option is used once.

BEHIND THE BUTTON 192W

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TAKES AS READ

a chain reaction needs the material held together long enough to grow · fission, neutron multiplication and chain reactions — taken as read

VERDICT 85W

Both concepts aim at the same goal. Bring subcritical material together fast enough that a chain reaction can grow before the assembly blows itself apart. They get there by completely different routes. A gun-type device drives one piece into another along one axis. That is a linear problem with a long history behind it. Implosion squeezes one piece inward with a converging wave, and nobody has built one. The two are not

merely different shapes. They create different organisations, different test programmes and different risks.

TAKEAWAY 8W

Weapon architecture determines the entire downstream engineering program.

2.4 Will it hold a chain, or will it not

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · BELT

SCENE 40W

A shipment has arrived and the vault wants it sorted before anything is stored together. Some of what is on this bench will sustain a chain reaction given enough of it. The rest will not, whatever you do with it.

GUIDE 50W

Two bins. A nuclide that fissions on slow neutrons will hold a chain given enough mass and the right geometry. One that only fissions on fast neutrons, or not at all, will not — however heavy the piece is. Sort on the nuclide, not on how dangerous the label looks.

ASK 15W

Send each item to the bin that says whether it can sustain a chain reaction.

BEHIND THE BUTTON 77W

Why the distinction is the whole of storage policy. Two pieces that each hold no chain can hold one together, so the rule about what may sit next to what is written from this sort. Nothing about mass or radioactivity substitutes for it.

Why radioactive and fissile are different words. Cobalt-60 will make a counter scream across the room and will never sustain anything. Natural uranium is barely active and contains the isotope everything here is about.

TAKES AS READ

some heavy nuclei split when they absorb a neutron · fission, neutron multiplication and chain reactions — taken as read

VERDICT 153W

A chain reaction needs a nuclide that fissions when it absorbs a neutron and then releases enough neutrons to do it again. Uranium-235 and plutonium-239 do that with slow neutrons, so a large enough piece in the right shape sustains itself. Uranium-238 fissions only on fast neutrons and absorbs the slow ones, so it damps a chain rather than carrying one — which is why natural uranium is nine-tenths a moderator by mass. Everything else on this bench is a different property entirely. Cobalt-60 and radium are intensely radioactive and cannot sustain anything. Beryllium and graphite are not radioactive at all and change everything, because they reflect and slow neutrons, which is why they are stored under the same rule. That is the point of sorting this way rather than by how loud a counter gets: the storage policy is written about what can happen when two pieces are put near each other.

TAKEAWAY 16W

What sustains a chain is decided by the nuclide, not by how radioactive a thing is.

END OF THE STAGE 20W

Mass defect sets the nuclear energy scale, but that scale becomes useful only after instruments and engineering goals are calibrated.

Follow radioactivity through the laboratory

BEFORE THE STAGE — FOLLOW

Groves walks the site the way he walks it

Groves, the commanding general, inspects on foot, fast, and stops where he stops for reasons he does not explain. Stay with him. What he looks at is the fastest available education in what this project is actually short of, and dropping behind means missing the half sentence in which he says it.

PLAN CARD 106W

July 1943, and 3 groups have reported yields that cannot all be right. Seaborg's chemistry says one thing, and Woods's counters another. The sample decays while the chemistry is still running, so the amount at the end is not the amount you started with. Today you handle decay as arithmetic, follow an element through a separation, and measure what a counter reads with nothing in front of it. Woods will not report a number until she has measured that, which is why hers is the count the others are being checked against. Without a background count taken first, the number on its own means very little.

BRIEFING 28W

A radioactive sample changes while it is being prepared and measured. Follow the same evidence chain from the decay law, through tracer chemistry, to a defensible background measurement.

WHAT YOU DECIDE 13W

Treat decay as a quantitative process that must survive chemistry and background measurement.

3.1 Radioactive decay law

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 37W

A tracer sample has been sitting through a six-hour shift and somebody needs to know what is left in it. Every count taken here is a count of something that has been decaying since it was made.

GUIDE 50W

Four numbers, and two of them are counts of half-lives that differ by a factor of ten. Ask of each whether it is a population, a fraction, or a number of halvings. Six hours of two-hour halvings is not thirty of them. And each halving multiplies by the same fraction.

ASK 20W

Estimate how many nuclei remain after the shift, then use the half-life to estimate the sample's activity at that moment.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a half-life is the time for half of a large population to decay · radioactive decay modes: alpha, beta, gamma — taken as read · exponential decay and logarithms — taken as read

VERDICT 73W

Decay is random for one nucleus and predictable for a large population. 6 hours is 3 two-hour half-lives, so 10,000 nuclei fall to 1 eighth, or about 1,250. Half-life also fixes the decay constant: $\lambda = \ln 2 / t_{1/2}$. For a two-hour half-life, λ is about $9.6 \times 10^{-5} \text{ s}^{-1}$. Activity is then $A = \lambda N$, so the remaining sample produces about 0.12 decays each second. Population and count rate are linked by the same decay law.

TAKEAWAY 19W

Half-life fixes both how a population falls with time and, through the decay constant, its activity at any moment.

3.2 Radiochemical tracers

CHEMISTRY & METALLURGY · A PERSON · BALLPARK

SCENE 30W

Chemistry and Metallurgy is running separations on quantities too small to weigh and needs to know how much is being lost. Every percent lost is weeks of production at Hanford.

GUIDE 50W

Four numbers, in two pairs, and only one pair belongs to this separation. Ask of each whether it is what went in or what came out. And note what makes a raw ratio dishonest here. The tracer decayed throughout, so both counts have to be corrected to the same moment.

ASK 4W

Estimate the chemical yield.

BEHIND THE BUTTON 106W

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TAKES AS READ

a radioactive sample announces itself through its own radiation

VERDICT 81W

A tracer is a small amount of radioactive material that travels with the chemistry and reports where it went. That turns an invisible loss into a measurable fraction: yield is what came out over what went in. The correction that makes it honest is decay — the tracer has been decaying throughout the separation, so a raw count afterwards understates the recovery and flatters the loss. Both counts have to be corrected to the same moment before the ratio means anything.

TAKEAWAY 10W

A tracer turns invisible material loss into a measurable fraction.

3.3 Find what carries the background

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · CONTROL

SCENE 36W

A weak sample has been counting for an hour and the question is whether it is emitting anything at all. This is where an experimental group either establishes a real signal or convinces itself of one.

GUIDE 75W

The number you are watching is the counting rate with no sample in the station, and the excess is how much of it cannot be accounted for. A change has told you something only when it moves that excess by more than the rate wanders on its own between runs. Several parts of the station changed when it was built, so choose one, run it, and put it back. Name the part the excess follows.

ASK 9W

Which part of the counting station carries the excess?

BEHIND THE BUTTON 178W

What a background is. A detector counts things whether or not you gave it anything to count: cosmic rays, activity in the concrete, contamination on the housing, noise in the electronics. The rate with no sample in place is the sum of all of those, and a weak sample is only detectable to the extent that its own counts stand above that sum.

Why one change at a time. Change two things and a response tells you only that one of them mattered. The station has a shield, a room and a detector, and each of them can carry background for its own reason, so the only reading that names a part is the one where nothing else moved.

Why putting it back matters. A rate that drops when you shield the detector and stays down when you unshield it was falling for some other reason — drift, a decaying source, somebody switching off a pump down the corridor. A cause has to follow the change in both directions, and that is the whole of what reversal tests.

TAKES AS READ

background is measured with no sample present · one changed variable can be tested against a stable baseline

VERDICT 78W

The sample cannot explain a count excess that remains after the sample is removed. Extra shielding cannot explain it when more lead changes nothing. The decisive comparison is hardware: the original detector still runs high in a clean room, while an identical clean detector in the same room returns to the normal background. Put the original detector back and the excess returns. That reversal makes contamination of the detector or holder a causal result rather than a correlation.

TAKEAWAY 19W

A suspected cause is strongest when changing only that cause moves the reading and reversing the change restores it.

3.4 Two weapon architectures at Los Alamos

ORDNANCE & ENGINEERING · A ROOM · CALLBACK · PROTOCOL

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GUIDE 49W

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ASK 16W

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TAKES AS READ

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VERDICT 85W

Both concepts aim at the same goal. Bring subcritical material together fast enough that a chain reaction can grow before the assembly blows itself apart. They get there by completely different routes. A gun-type device drives one piece into another along one axis. That is a linear problem with a long history behind it. Implosion squeezes one piece inward with a converging wave, and nobody has built one. The two are not

merely different shapes. They create different organisations, different test programmes and different risks.

TAKEAWAY 8W

Weapon architecture determines the entire downstream engineering program.

END OF THE STAGE 17W

A count rate is not automatically a decay rate; time dependence, chemical recovery, and background all matter.

Measure interactions, not impressions

PLAN CARD 125W

October 1943. Two groups have a difference in their counts that everybody believes in. Wu has run the numbers and the statistics say it is nothing, and the head of the Theoretical Division, Hans Bethe, has asked for both to be recalculated by somebody who did not do them the first time. The programme is about to allocate material on the strength of it. Today you work with the number that says how likely an interaction is. You decide when a difference in counts is real. And you check that a chemical yield is what it claims. A yield figure means nothing without a purity figure beside it — which is the question Wu asks first and the one the reporting group did not answer.

BRIEFING 36W

The laboratory now needs probabilities rather than anecdotes. Estimate how often an interaction occurs, determine whether a count difference is statistically meaningful, and verify that the chemistry did not lose an unknown fraction of the sample.

WHAT YOU DECIDE 14W

Learn how cross sections, counting statistics, and analytical yield turn observations into quantitative evidence.

4.1 Reaction cross sections

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 30W

The Theoretical Division needs a reaction rate and has a cross section written in barns. Nearly every number this laboratory argues about is built out of the multiplication that follows.

GUIDE 53W

Four numbers, and two of them are areas thirty powers of ten apart. Ask of each whether it is a density, an area, or a bare one. Check the units before choosing, because a cross section is an area. One parameter comes out of the first step and then does all three jobs.

ASK 22W

Estimate the macroscopic cross section, the mean free path, and the fraction of an uncollided beam left after one mean free path.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a cross section is expressed as an area · fission, neutron multiplication and chain reactions — taken as read

VERDICT 76W

A microscopic cross section is an interaction probability written as an area. Multiply it by the number of targets per volume and you get the macroscopic cross section Σ , an inverse length. Its inverse is the mean free path. The same Σ also controls attenuation: $I/I_0 = e^{-\Sigma x}$. After 1 mean free path, $x = 1/\Sigma$, so the exponent is minus 1 and about 37% of an uncollided beam remains. One parameter connects all three ideas.

TAKEAWAY 22W

Number density turns a microscopic cross section into an interaction probability per length, which sets both mean free path and exponential attenuation.

4.2 Poisson counting statistics

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · BALLPARK

SCENE 31W

A run has finished with 400 net counts and somebody wants to know how well the result is known. Beam time, detector time, material, and shift hours are all scarce here.

GUIDE 47W

Four numbers, and three of them are count totals. Only one belongs to this run. Ask of each whether it is the numerator or the total. And note how slowly this improves. Halving the uncertainty needs four times the counts, which makes run length a planning decision.

ASK 5W

Estimate the fractional statistical uncertainty.

BEHIND THE BUTTON 106W

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TAKES AS READ

radioactive decays arrive at random · descriptive statistics: mean, spread, standard deviation — taken as read

VERDICT 72W

A repeat of the same run would not give exactly the same total. For Poisson counting, the typical fluctuation is about the square root of the count, so the fractional uncertainty is $1/\sqrt{N}$. 400 counts gives 5%. To cut that in half, to 2.5%, you need 4 times as many counts. That slow square-root improvement is why counting statistics are a planning tool: longer runs buy precision, but they buy it expensively.

TAKEAWAY 13W

Four times as many counts are needed to cut statistical uncertainty in half.

4.3 Close the material balance

CHEMISTRY & METALLURGY · A ROOM · BALANCE

SCENE 38W

A separation batch has come back and the numbers do not close. Downstream, somebody is going to cast this material assuming the report meant what they thought it meant, so Chemistry has to close the ledger before release.

GUIDE 49W

Read every stream first, because reading costs nothing. Then count only the masses that belong to the same material ledger. The batch does not close, and the missing mass has gone somewhere real — an uncounted route rather than an arithmetic slip. Report the total when the ledger closes.

ASK 7W

Which uncounted route closes the material balance?

BEHIND THE BUTTON 130W

Why a balance has to close. Material is conserved, so a batch that does not add up is a statement that something has not been measured, not that some of it stopped existing. Every unclosed ledger is a location nobody has looked at yet.

Where material actually goes. Holdup in the equipment, residue on filters and vessel walls, a side stream taken for analysis, losses to a scrubber. Each is ordinary and each is invisible in the streams somebody thought to weigh.

Why it matters downstream. Somebody is going to cast this material assuming the report means what they think it means. In this programme, an assay that is quietly wrong about how much is present is the kind of error that ends with a criticality rather than a memo.

TAKES AS READ

percent of input material can be added across mutually exclusive output routes · purity and recovered quantity are different measurements

VERDICT 69W

The input is 100% of the target material. Recorded product and aqueous output account for only 93.9%, so 6.1% is still missing. The 91% purity result answers a different question and cannot be added to that ledger. A wall swab finds 5.8% of the input on the vessel surface. Add that loss route and the balance closes to 99.7%, inside rounding. Yield, purity, and material balance are different claims.

TAKEAWAY 19W

A purity result cannot substitute for a closed material balance; losses outside measured process streams must still be counted.

4.4 Two weapon architectures at Los Alamos

ORDNANCE & ENGINEERING · A ROOM · CALLBACK · PROTOCOL

SCENE 41W

Oppenheimer has asked Ordnance for a recommendation before the next program meeting. Two assembly concepts are still alive, but choosing between them commits people, buildings, and months of work. The room is waiting for a decision that will reshape the Hill.

GUIDE 49W

Four situations and four responses, and both designs chase the same goal. Ask of each whether it is about the geometry of the assembly or about its timing. One route drives two pieces along a line. One squeezes inward, and nobody has built one. A measurement decides between them.

ASK 16W

Match each situation to the best action, control, document, or interpretation. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a chain reaction needs the material held together long enough to grow · fission, neutron multiplication and chain reactions — taken as read

VERDICT 85W

Both concepts aim at the same goal. Bring subcritical material together fast enough that a chain reaction can grow before the assembly blows itself apart. They get there by completely different routes. A gun-type device drives one piece into another along one axis. That is a linear problem with a long history behind it. Implosion squeezes one piece inward with a converging wave, and nobody has built one. The two are not

merely different shapes. They create different organisations, different test programmes and different risks.

TAKEAWAY 8W

Weapon architecture determines the entire downstream engineering program.

END OF THE STAGE 14W

Reliable nuclear evidence combines interaction probability, statistical uncertainty, and a measured recovery or efficiency.

Track neutrons through matter

PLAN CARD 102W

April 1944, and the first reactor plutonium has arrived on the mesa. It is not the material the gun design was built around, and Bethe has spent a week with the counting data saying so quietly. His numbers have been on Oppenheimer's desk since Tuesday and nothing has been decided. Today you follow what happens to neutrons as they move through matter. You work out how to detect something with no charge. And you face what this material means for the second design path. The gun programme has a year of work in it, and 11 people who do not know yet.

BRIEFING 27W

Neutrons change energy, direction, and detectability as they move through matter. Use that foundation to understand why the plutonium program could not simply copy the first architecture.

WHAT YOU DECIDE 14W

Connect moderation, neutron detection, and the historical reason plutonium forced a different design path.

5.1 Neutron energy and moderation

THEORY & CALCULATIONS • A ROOM • PROTOCOL

SCENE 33W

A neutron leaving a fission event carries a great deal of energy, and what happens next depends entirely on what it hits. Confusing the possibilities is how a shielding calculation goes quietly wrong.

GUIDE 49W

Four things that can happen to a neutron, and four processes. Pair them by asking two questions. Is the neutron still there afterwards? And how much energy did it give up? A heavy target barely recoils, so it takes almost nothing away. One of these removes the neutron entirely.

ASK 18W

Match each information need or situation to the best action, instrument, or control. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a light ball loses more energy hitting something its own size • fission, neutron multiplication and chain reactions — taken as read

VERDICT 87W

Slowing, redirecting and absorbing are three different processes with three different consequences. In elastic scattering the neutron transfers some energy and continues as a neutron — it is still in the system. In absorption it is captured into a compound nucleus and is gone. Scattering off a heavy nucleus changes direction and gives up very little energy, because a heavy target barely recoils. Scattering off light nuclei is what slows neutrons efficiently, because a light nucleus can take a large share of the energy in one collision.

TAKEAWAY 8W

Energy management and neutron removal are not interchangeable.

5.2 Are the neutrons missing, or the counts?

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · TRACE

SCENE 38W

A neutron counter is reading well below prediction and nobody yet knows whether the physics or the instrument is at fault. The measurement about to be built on this counter is one the whole plutonium programme depends on.

GUIDE 51W

Open each reading to see what it was computed from. Keep the ones that could still see neutrons arriving without passing through the discriminator, and untick the rest. Then name the dependency the failed readouts share. Both halves count: discarding an independent measurement costs as much as keeping a dependent one.

ASK 12W

Are neutrons failing to arrive, or is the counting chain discarding them?

BEHIND THE BUTTON 127W

Why the two explanations look identical on the console. Fewer neutrons arriving and fewer neutrons being counted produce the same low number. Nothing about the number itself separates them, which is why the answer has to come from what each channel depended on rather than from its value.

What the counting chain contains. A detector, a discriminator that decides whether a pulse is large enough to count, an amplifier, a scaler. A threshold set too high silently rejects real events, and every channel downstream of it reports the same shortfall convincingly.

Why an independent channel is worth so much here. An activation foil or a geometry check sees neutrons arriving without the electronics deciding anything. One such measurement settles a question no number of counter runs can.

TAKES AS READ

a neutron detector converts an interaction into an electrical pulse · a discriminator rejects pulses below a set threshold

VERDICT 79W

The main counter is low, but that result depends on every stage of the counting chain. The activation foil does not: it sees the expected neutron exposure without using the discriminator electronics. Geometry also behaves as expected under the inverse-square check. Meanwhile the pulse-height distribution piles up just below the discriminator threshold. Those observations can all be true together only if neutrons are arriving and real detector pulses are being rejected. Agreement among dependent readouts is not independent evidence.

TAKEAWAY 17W

Independent evidence can separate a missing physical signal from a failure in the instrument that reports it.

5.3 Why plutonium required implosion

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · PROTOCOL

SCENE 41W

Measurements on reactor-produced plutonium have come back from the counters and Ordnance has been called in. The result changes not one component but the direction of the entire programme, and the reason has to be understood before the reorganization makes sense.

GUIDE 50W

Four situations and four responses. Ask of each whether it is about the material or about the engineering. This material emits neutrons on its own, at a rate nobody chose. So the clock every assembly has to beat is set by what comes out unprompted, not by a design decision.

ASK 16W

Match each situation to the best action, control, document, or interpretation. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a chain reaction started early releases energy early · pressure, temperature and equations of state — taken as read

VERDICT 88W

Reactor-produced plutonium contains isotopes that emit neutrons on their own. The material supplies its own stray neutrons, at a rate nobody chose. A slow assembly spends longer in the state where one of those could start the chain. A chain that starts early releases enough energy to blow the pieces apart before most of the material has reacted. That is pre-initiation. It is a property of the material, not a

defect better workmanship can fix. Implosion is the answer: squeeze from all sides, and cut the vulnerable time.

TAKEAWAY 8W

Material physics can determine the required system architecture.

5.4 Isotopes and chemical identity

CHEMISTRY & METALLURGY · A ROOM · CALLBACK · PROTOCOL

SCENE 33W

Chemistry and Metallurgy is being asked what it can and cannot deliver, and the answer turns on one fact about isotopes. Everything shipped here has both a chemical form and an isotopic composition.

GUIDE 48W

Four questions and four answers, and three of them say yes. Ask of each whether the property lives in the electrons or in the nucleus. Bonding is electrons, and extra neutrons barely disturb them. Stability is nuclear. So ordinary chemistry has almost no handle here, and mass does.

ASK 18W

Match each information need or situation to the best action, instrument, or control. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

chemical bonding is done by electrons

VERDICT 84W

Isotopes of an element carry the same proton count and nearly the same electrons, so they form the same compounds and answer to the same reagents. Ordinary chemistry therefore has almost no handle for separating them. Nuclear stability is different: it is set inside the nucleus, and half-lives can differ enormously between isotopes of one element. Mass spectrometry sorts by mass-to-charge ratio, responding to the difference chemistry mostly ignores. That is why isotopic separation is a physical problem as well as a chemical one.

TAKEAWAY 10W

Materials work must track both chemical form and isotopic composition.

END OF THE STAGE 9W

Neutron behavior links microscopic interactions to system-level design choices.

Recognize the design pivot

BEFORE THE STAGE — HUNT

Five detectors signed out, and the log accounts for three

Every detector on this mesa is signed out against a name, and two of them are against nobody. A detector nobody can account for is a count somebody may have taken and not written down, and the whole measurement rests on knowing which instrument read what.

PLAN CARD 95W

July 1944. A process nobody had to care about last spring is about to end a year of work in a fortnight, and Oppenheimer has to reorganise the laboratory around the consequence. Today you measure the neutron background the new material produces, follow the argument that killed the earlier design, and take the programme to a different one. That is the entire argument, and it is made of a counting rate on one bench in one room. Half the site has to be told why the work they have done for a year is over.

BRIEFING 31W

A new material property changes the program. First identify the background process, then see why the earlier approach became unacceptable, and finally translate the problem into the idea of inward compression.

WHAT YOU DECIDE 15W

Use neutron-background evidence to understand why one path was abandoned and inward compression became central.

6.1 Spontaneous fission and neutron backgrounds

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 34W

A sample on the bench is emitting neutrons all by itself, at a rate the counters can measure. 2 assembly schemes are on the table, and 1 takes 1000 times longer than the other.

GUIDE 50W

Four numbers: two rates and two windows. Ask of each which assembly it belongs to. The rate is the same either way and the window is not. A thousandth of a second gives a thousand times as many chances as a millionth. The material stays identical while the verdict changes.

ASK 14W

Estimate the neutrons expected during the slow assembly, then compare with the fast one.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

some heavy nuclei split without being hit by anything · radioactive decay modes: alpha, beta, gamma — taken as read

VERDICT 76W

A spontaneous-fission source gives a steady random background of neutrons.

Whether that background matters depends on the time spent vulnerable to an early neutron. The expected number in that window is simply rate times time. A process lasting a thousandth of a second sees 1000 times as many chances as 1 lasting a millionth. The material and source rate can stay the same while the timing changes the verdict completely. That is why the arithmetic matters.

TAKEAWAY 19W

The same background source term is negligible or decisive depending on how long the system is exposed to it.

6.2 Why Thin Man was abandoned

ORDNANCE & ENGINEERING · A PERSON · SEQUENCE

SCENE 34W

Thin Man was the plutonium gun design and much of the site's effort was pointed at it. Then the counters returned a number the concept could not survive, and within weeks the laboratory reorganised.

GUIDE 50W

These four are one chain, from a measurement to a decision that cost a year. Ask of each card what has to be known before it. Nothing is unacceptable before it is understood, and nothing is understood before it is measured. The reorganisation is the consequence rather than the argument.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a measurement can rule out a design without anybody being wrong · pressure, temperature and equations of state — taken as read

VERDICT 81W

This is what a measurement invalidating a development path looks like from inside: not a debate about opinion, but a chain from one new fact to a decision that costs a year. The unexpectedly high spontaneous-neutron background in reactor plutonium comes first, because it is the new evidence. That makes a slow assembly vulnerable to starting early. That vulnerability makes the plutonium gun unacceptable. And only then does the effort move to implosion, which is a decision rather than a fact.

TAKEAWAY 12W

In complex engineering, one new measurement can invalidate an entire development path.

6.3 From outward detonation to inward compression

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · SEQUENCE

SCENE 31W

Ordinary high explosive detonates outward from wherever it was lit, and every intuition anybody on the Hill has comes from that. Implosion needs the opposite, and nobody has done it before.

GUIDE 50W

These four are one chain, and the problem is convergence rather than power. Ask of each card what has to happen first. One point of initiation gives one outward wave. Waves have to exist before they can be redirected. And the density change is the point of all of it.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

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TAKES AS READ

a detonation travels outward from where it starts · pressure, temperature and equations of state — taken as read

VERDICT 91W

The central problem is controlled convergence rather than raw explosive power. The outer explosive system is initiated at multiple controlled points, because one point gives one outward wave and nothing else. Those waves are then shaped and redirected toward the centre — this is the step that has no precedent. The resulting inward front drives through the inert layers and the assembly. And the density has to rise fast enough to change the neutron balance before expansion begins, which is what makes speed part of the physics rather than a preference.

TAKEAWAY 11W

The central challenge is controlled convergence, not simply a larger explosion.

6.4 Ionization and detector signals

EXPERIMENTAL PHYSICS & DIAGNOSTICS • EXPERIMENTAL PHYSICS • A ROOM •
CALLBACK • SEQUENCE

SCENE 36W

The Physics Division cannot see a single neutron or alpha particle, and neither can anybody else. Everything the Hill will claim about cross sections, backgrounds and yields rests on a chain of conversions working in order.

GUIDE 51W

These four are one chain, and a detector never measures the radiation itself. Ask of each card what has to exist before it. Nothing is amplified before something is set free. And nothing is set free before energy is deposited. That middle step decides what the instrument can see at all.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

radiation deposits energy in whatever it passes through • decay constant, half-life and activity — taken as read

VERDICT 86W

A detector measures the consequences of an interaction, never the radiation itself. Energy has to be deposited in the sensitive material first — nothing downstream exists until that happens. That deposit sets free charge or light, which is the transducer step and the one that decides what the instrument is sensitive to at all. Electronics then collect and amplify a response too small to handle raw. The data system digitises and stores a pulse last. Get the chain wrong and you are measuring your own apparatus.

TAKEAWAY 11W

A detector measures the consequences of an interaction, not radiation directly.

Programs must change architecture when new evidence invalidates an original assumption.

Understand matter under compression

PLAN CARD 116W

September 1944, and compression is now the whole programme. Lilli Hornig, the chemist casting the metal at the centre of it, has found that it changes density under conditions the design will actually meet. Nobody wanted that, and the mathematician John von Neumann wants the shock tables redone with it in. Today you work with the relationship between pressure, volume and temperature, read what a phase diagram is telling you, and understand what shaping a detonation wave achieves. Turning a wave that spreads outward into one that converges inward is von Neumann's problem before it is the explosives division's, under George Kistiakowsky: it is done in numbers first, and only then does X Division cut anything.

BRIEFING 30W

Compression depends on how materials respond under pressure and how waves cross boundaries. Build the physical picture from bulk matter, to material phase behavior, to the purpose of wave shaping.

WHAT YOU DECIDE 16W

Connect equations of state, material phases, and wave shaping without jumping directly to a full system.

7.1 Equations of state

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 32W

The Theoretical Division is calculating work done on a material at pressures nothing in the laboratory has ever measured. An error in the units looks exactly like an error in the physics.

GUIDE 46W

Four numbers: two pressures and two volumes, each pair a hundredfold apart. Ask of each whether it describes this regime or handbook conditions. Work is a pressure times a volume change, so an error in either is an error of the same size in the answer.

ASK 4W

Estimate the work scale.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

pressure times a volume change is work · units and conversions: MeV, u, barns, scientific notation — taken as read

VERDICT 88W

An equation of state ties pressure, density, temperature and internal energy together for a substance. The familiar handbook values were all taken at room conditions. They describe a completely different regime from the one implosion makes. That is why this work is unforgiving in two directions at once. The thermodynamics has to be right. So does every unit conversion carrying it across many powers of ten. Pressure in pascals times a volume change in cubic metres gives joules, and the powers of ten do most of the work.

TAKEAWAY 10W

Extreme-state calculations require both thermodynamic relationships and careful unit scaling.

7.2 Phase diagrams

CHEMISTRY & METALLURGY · A PERSON · DIAGNOSIS

SCENE 35W

A metallurgist has a microstructure under the microscope that the phase diagram says should not be there. Everything the laboratory does with this metal is an argument with a diagram drawn from very little data.

GUIDE 53W

A phase diagram is drawn for one composition. Ask of each candidate whether it questions the map, or what the map was drawn for. Plutonium has several solid phases and large volume changes between them, so a crossed boundary is a real effect. But check you still have the material the diagram describes.

ASK 19W

Read the whole reading panel, including the calm readings, then select the one cause that fits all of it.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a metal can exist in more than one solid form · units and conversions: MeV, u, barns, scientific notation — taken as read

VERDICT 85W

A phase diagram maps which solid and liquid phases are stable at each temperature and composition. Crossing a boundary can melt a material, throw out a second phase, or change how it behaves under load. Plutonium is notorious for it. Several solid phases, and large volume changes between them. But a diagram describes one composition. So before doubting the map, check you still have the material it was drawn for. A trace impurity is enough to hold open a phase that should not be there.

TAKEAWAY 9W

Composition and temperature jointly determine which phases are possible.

7.3 What an explosive lens does

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM ·
PROTOCOL

SCENE 33W

The lens programme has been given the problem of shaping a detonation front, and the name is a deliberate analogy. Every part of that analogy has to be earned against real firing data.

GUIDE 45W

Four questions and four ways to answer them. Pair them by asking what each measurement isolates. One material alone? Two acting together? The joints between segments? Or the whole wave around the assembly? Manufacturing is physics here, because a gap distorts the wave crossing it.

ASK 14W

Match each lens-program question to the measurement or engineering action that can answer it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a lens bends light by slowing it in some places more than others · pressure,
temperature and equations of state — taken as read

VERDICT 83W

An explosive lens puts together explosives with different detonation speeds. A front crossing the boundary between them is reshaped, and that speed contrast is the working principle. The boundaries are shaped so faster and slower paths arrive together, as one inward-curving front. Manufacturing then matters as physics. A gap or a property change at a segment boundary distorts the wave crossing it. And

symmetry has to be measured rather than assumed, because detonation is violent and nonlinear in a way light is not.

TAKEAWAY 12W

A conceptual analogy must still be tested against the actual nonlinear physics.

7.4 One reference line is not a calibration

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM ·
CALLBACK · DEGENERACY

SCENE 43W

Two Physics groups bring spectra from different instruments, and the peaks refuse to line up. Each group trusts its own axis. An afternoon comparison meeting is approaching, and somebody has to decide whether the disagreement belongs to the samples or to the instruments.

GUIDE 49W

Two controls, gain and offset, slide along a family of calibrations that all put one reference line where it belongs. Find out how wide that family is before deciding anything. Then bring in the second reference line, which does not permit the same trade, and watch the family collapse.

ASK 10W

Which gain and offset are fixed by both reference lines?

BEHIND THE BUTTON 151W

Why one line cannot calibrate an axis. A linear axis has two unknowns: how much a step is worth, and where zero sits. One known peak gives one equation. Every gain paired with the offset that puts that peak in the right place fits equally well, which is exactly the family the controls slide along.

Why the second line is not a repeat. It constrains the same two unknowns through a different combination, so the pairs that satisfied the first no longer satisfy this one. Where the two constraints cross is the calibration, and it is why standards come with more than one line.

What is at stake this afternoon. If the axes can be reconciled, the spectra agree and the disagreement was instrumental. If they cannot, the difference is in the samples. Both groups trusting their own axis is how a meeting spends an afternoon arguing about the wrong thing.

TAKES AS READ

a straight-line calibration can have both a slope and an offset · one equation cannot determine two unknowns

VERDICT 70W

A straight-line calibration has two unknowns: gain and offset. One known line gives only one equation, so many gain-and-offset pairs reproduce it exactly. That is not noise; it is missing information. A second line at a different energy supplies a second equation. The two allowed loci cross at one point, fixing both parameters. This is why a convincing-looking one-point calibration can still place every unknown peak at the wrong energy.

TAKEAWAY 18W

One calibration point cannot determine both a slope and an offset; a second independent point breaks that degeneracy.

END OF THE STAGE 14W

A compression concept is only credible when material response and wave propagation support it.

Make symmetry measurable

PLAN CARD 110W

October 1944. “Nearly simultaneous” has been in every memo for two months. Nobody has written down what it means in microseconds. That is why Kistiakowsky's division and Bethe's have been arguing past each other since August. Today you turn it into a number, calibrate the channels that measure it, and give ordnance a requirement they can build to. Kistiakowsky has four recording channels and no record of when any of them was last checked against another. The memos will go on saying nearly simultaneous until somebody writes a tolerance beside it. If the instruments disagree about when things happened, the timing you report describes the instruments rather than the shot.

BRIEFING 25W

“Nearly simultaneous” is not a measurement. Define a symmetry metric, calibrate the timing channels that observe it, and see why simultaneity became a system requirement.

WHAT YOU DECIDE 14W

Turn a qualitative demand for symmetry into calibrated timing evidence and an engineering requirement.

8.1 How many half-lives have passed

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · BALLPARK

SCENE 34W

A sample counted at 8,000 counts a minute reads 500 today, and the notebook does not say how long ago the first count was taken. The half-life is on the chart at 12 hours.

GUIDE 52W

Six numbers, and three of them are steps of the working: a ratio, a count of halvings, and a base. Ask of each whether the relationship needs it or produces it. And note the habit. Work in ratios rather than differences, because a drop of 7,500 counts means nothing on its own.

ASK 8W

Estimate the elapsed time since the first count.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a half-life is the time for half of a sample to decay

VERDICT 81W

Every half-life halves the count rate, so the elapsed time is the number of halvings times the half-life. 8000 to 500 is a factor of 32 is 2 to the fourth: 4 half-lives, or 48 hours. Written as a logarithm it is log base two of the ratio, which is what a semi-log plot shows as a straight line. The useful habit is to work in ratios rather than differences — the difference of 7,500 counts means nothing on its own.

TAKEAWAY 16W

Exponential decay is counted in half-lives, and logarithms are how you get the number of them.

8.2 Find where timing jitter enters

EXPERIMENTAL PHYSICS & DIAGNOSTICS • EXPERIMENTAL PHYSICS • A ROOM • CONTROL

SCENE 30W

A timing array is disagreeing with itself, and until that is settled no symmetry claim from it means anything. The Hill is about to spend months measuring differences of nanoseconds.

GUIDE 69W

The number you are watching is the spread in arrival times between channels, and the live path gives about fourteen nanoseconds of it. Substituting a reference part asks a specific question: if that part were perfect, how much spread would be left? A substitution that leaves the spread where it was has cleared that part. Run one at a time, restore it, and name the stage the spread follows.

ASK 14W

Which part of the live timing path is responsible for the broad arrival-time spread?

BEHIND THE BUTTON 165W

What the spread is made of. Every stage between the particle and the recorded time adds its own variation: how the detector forms a pulse, how the cable carries it, how the discriminator decides when the pulse crossed a threshold. Those add in quadrature, so the largest one dominates and the others are nearly invisible until it is removed.

Why substitution rather than adjustment. A reference pulse injected at the electronics input is a known, jitter-free signal. Feeding it in replaces one stage with something whose contribution is negligible, so what remains is everything else. That makes the measurement subtractive: the drop when a stage is bypassed is that stage's own contribution.

What is riding on it. Months of measurement here are differences of nanoseconds, so a timing path that adds fourteen of them is not a detail. Until the array agrees with itself, no asymmetry it reports can be separated from its own jitter, which is why this is settled before the physics starts.

TAKES AS READ

a fixed delay shifts a distribution while random jitter broadens it • a controlled substitution changes one part of a measurement chain at a time

VERDICT 81W

A useful control changes one suspected part of the measurement chain while leaving the rest in place. The live timing path shows a 14 ns spread. Equalized cables or a replacement discriminator leave nearly the same broad distribution, so neither explains it. Identical pulses injected at the electronics inputs collapse the spread to about 1 ns. When the live detector path is restored, the broad spread returns. The changing element is therefore the live pulse-formation stage upstream of the injection point.

TAKEAWAY 21W

A controlled substitution localizes an instrument error only when one change removes the effect and reversing that change brings it back.

8.3 Why simultaneity became an engineering requirement

ORDNANCE & ENGINEERING · A ROOM · BALLPARK

SCENE 41W

Several initiation channels have to act so nearly together that their separate wavefronts merge into one coordinated event. On the timing bench, path length is one measurable source of delay, and the conversion from length to time has to be explicit.

GUIDE 53W

Four numbers, and two of them are signal speeds a hundred times apart. Ask of each whether it could carry a pulse down a cable. And note what the answer is about. A path difference is an arrival difference, so a small delay makes an asymmetric implosion rather than a slightly worse one.

ASK 4W

Estimate the delay difference.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a signal travels along a cable at a finite speed

VERDICT 86W

A small delay at one point does not produce a slightly worse implosion; it produces an asymmetric one. So simultaneity stops being a matter of setting clocks and becomes a physical interface problem: a difference in cable path length is a difference in arrival time, at the speed the signal travels. Divide the path mismatch by that speed and the answer is in nanoseconds. Cables, capacitors, switches and firing units all contribute time, and all of it has to be accounted for rather than assumed small.

TAKEAWAY 11W

Timing synchronization is a physical interface, not merely a clock-setting problem.

8.4 Just critical, for as long as it takes

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A PERSON · HOLD

SCENE 40W

The pile has to sit exactly critical while a foil is irradiated on the other side of the shield. The control rod is the only thing that answers it, and everything else in the room is quietly changing the balance.

GUIDE 65W

Hold the pile at the neutron level on the meter, inside the band. The band narrows as the run goes on, because the foil's exposure has to be known to a few per cent and every excursion has to be accounted for afterwards. The control rod is your control, and every change in the pile is a change in the rate rather than a step.

ASK 12W

Hold the pile at the neutron level while the foil is irradiated.

BEHIND THE BUTTON 123W

Why a pile does not sit still. Exactly critical means each generation produces the same number of neutrons as the last. A little above and the population grows without limit; a little below and it dies away. There is no stable place except the one you hold.

What is moving it. The graphite warms and moderates a little differently. A xenon product builds up and absorbs neutrons. Somebody walks past the reflector. Each is a slow push in one direction, and it does not stop.

Why the rod is set rather than tapped. A rod moved and returned changes the population and leaves it changed. Holding the level means finding the rod position where growth and loss balance under whatever is currently pushing.

TAKES AS READ

a chain reaction can grow, hold steady or die away · radioactive decay modes: alpha, beta, gamma — taken as read

VERDICT 151W

Exactly critical is not a place a pile rests. Each generation either produces more neutrons than the last or fewer, so the level grows or decays and the only steady state is the one an operator makes continuously. That is why the rod is set to a position and held rather than tapped: every push here is a rate, and a rod returned to where it was leaves the population somewhere new. The pushes themselves are the physics of the pile. Warming graphite moderates a little differently, so the multiplication drifts as the run heats up. Fission products build up and absorb neutrons, which is a slow, one-way pressure downward that will not stop when you stop watching. And the band narrows because the foil is the experiment: an exposure known to a few per cent is a result, and one that wandered is a foil somebody has to irradiate again.

TAKEAWAY 13W

A balance with no stable point is one somebody has to keep making.

8.5 The nucleus as a physical system

THEORY & CALCULATIONS · A ROOM · CALLBACK · PROTOCOL

SCENE 33W

Your first morning on the Hill. The Theoretical Division hands you four nuclide labels with no explanation and expects you to pull the right quantity out of each one before anybody calculates anything.

GUIDE 52W

Four things you need and four ways to get them. Pair them by asking what each symbol counts. One counts protons. One counts protons and neutrons together. So one of these answers is a subtraction rather than a symbol on the label. And a neutral atom's electron count follows from its protons.

ASK 18W

Match each information need or situation to the best action, instrument, or control. Each option is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an atom has a nucleus of protons and neutrons, with electrons around it

VERDICT 91W

Notation is only useful when each symbol is tied to a physical count. The atomic number Z is the proton count, and it fixes which element this is. The mass number A counts protons and neutrons together, so the neutron count is A minus Z — a subtraction, not a symbol on the label. Two atoms are isotopes of one element when Z is the same and A differs. And a neutral atom has as many electrons as protons, which is why Z decides the chemistry as well as the identity.

TAKEAWAY 11W

Nuclear reasoning starts by translating notation into particles and conserved quantities.

END OF THE STAGE 16W

A system requirement becomes testable only when it has a metric and a calibrated measurement chain.

Build evidence with real materials

BEFORE THE STAGE — CANVASS

Who is worried about the same thing you are

Somebody on this mesa has already thought about the problem in front of you and security means you will not be told who. Ask around the divisions until enough answers come back to say whether it has been solved, half solved, or never raised.

PLAN CARD 108W

December 1944. The thing itself cannot be tested, so everything the programme believes has to come from stand-ins, and the committees run by Robert Bacher, who heads the physics division, exist to force the interfaces between them. They meet on Thursdays, and every division arrives with a different definition of ready. Today you characterise how the real materials behave under load, decide what a mockup can and cannot tell you, and choose diagnostics for events too fast to watch. An event lasting microseconds leaves no witness at all unless the instrument was chosen and placed beforehand. That means high-speed imaging, and radiography, which sees through what light cannot.

BRIEFING 24W

The project needs evidence before a full integrated event. Characterize the materials, build representative non-nuclear hardware, and choose diagnostics that reveal fast internal behavior.

WHAT YOU DECIDE 14W

Use mechanical properties, inert mockups, and high-speed imaging to test a difficult system safely.

9.1 Mechanical properties

CHEMISTRY & METALLURGY • A ROOM • DIAGNOSIS

SCENE 39W

A component has failed in service and the failure analysis has landed on your bench. This programme cannot spare either the material or the weeks, so the question is what changed between the qualified coupon and this actual part.

GUIDE 55W

Four properties, and none of them is a synonym for another. Ask of each what it actually measures. Deflection under load? Where permanent bending begins? Resistance to being dented? Or energy absorbed before it breaks? A material can be hard and brittle, and that is how a part passes a test and fails in service.

ASK 19W

Read the whole reading panel, including the calm readings, then select the one cause that fits all of it.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a material can be strong in one sense and weak in another • units and conversions: MeV, u, barns, scientific notation — taken as read

VERDICT 85W

Four properties are involved and they are not synonyms. Elastic modulus is stiffness — how much it deflects under load. Yield strength is where permanent deformation begins. Hardness is resistance to indentation. Toughness is how much energy the material absorbs before it fractures. A material can be hard and brittle, or soft and very tough, so choosing the wrong one of these as 'strong' is how a part that passed every specification arrives cracked. A part is only as good as the property nobody inspected.

TAKEAWAY 11W

Strength, stiffness, hardness, and toughness should not be treated as synonyms.

9.2 Why the limit is written down

ORDNANCE & ENGINEERING · A PERSON · CHOICE

SCENE 38W

Two half-assemblies go on the same bench this afternoon. The procedure sets a mass limit and a spacing, both comfortably inside the calculated physical boundary, and the crew wants to know why the written margin is deliberately larger.

GUIDE 57W

Four arguments for a written limit. Ask of each what it assumes about the other terms. Criticality answers to mass, geometry, reflection and moderation, and any of those can move by accident. So a limit on mass alone cannot make a piece safe in isolation. And a written number does not depend on who is on shift.

ASK 17W

The written limit is well below the critical mass. What is the argument for keeping it there?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

criticality depends on geometry and surroundings as well as on mass

VERDICT 74W

Criticality depends on mass, geometry, reflection and moderation. Any of those can change by accident. A hand nearby reflects neutrons. A spilled solution moderates them. A second piece on the bench adds mass. The written limit sits far below the physical one, so one human error cannot reach it. It is written down so the margin does not depend on who is on shift. Two of this laboratory's fatal accidents were exactly that error.

TAKEAWAY 15W

Administrative limits create margin against combinations of mass, geometry, reflection, moderation, and ordinary handling error.

9.3 High-speed imaging and radiography

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · BALLPARK

SCENE 43W

On the analysis table are high-speed photographs from yesterday's implosion test. Beside them sit several radiographs of the opaque assembly. Each image is labeled with its exposure time. The notebook also lists estimated object speeds and the feature sizes the team wants measured.

GUIDE 47W

Four numbers, and two of them are exposure times a thousand apart. Ask of each which one this camera used. And note what the answer is: a limit the optics cannot beat. Anything moving during the exposure travels a real distance, and that distance is the blur.

ASK 4W

Estimate the blur distance.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a photograph records everything that happened while the shutter was open

VERDICT 80W

Anything moving during the exposure travels a real distance, and that distance becomes motion blur: speed times exposure time. 500 metres per second across 1 microsecond is half a millimetre. That is a resolution limit even if the optics are perfect. Exposure time, viewing angle, penetration, and timing all decide what an image can establish. Blur is especially useful because it is calculable before the shot, so the diagnostic can be designed around the feature size rather than judged afterwards.

TAKEAWAY 11W

Exposure time sets a direct spatial-blur limit for rapidly moving objects.

9.4 Mass defect and binding energy

THEORY & CALCULATIONS · A ROOM · CALLBACK · BALLPARK

SCENE 37W

A theorist at the blackboard wants a number before lunch: how tightly a medium-heavy nucleus is bound. The arithmetic is small enough for one line, but the energy scale is the reason this laboratory exists at all.

GUIDE 48W

Four numbers, and two of them are the same defect written a thousand times apart. Ask of each whether it is per nucleon or per nucleus. The conversion turns mass into energy, so it belongs beside the mass term. And work out the per-nucleon figure before the total.

ASK 11W

Estimate the total binding energy and the binding energy per nucleon.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

mass and energy are related by $E = mc^2$, with E the energy, m the mass and c the speed of light

VERDICT 88W

A bound nucleus weighs less than the loose protons and neutrons that went into it. That missing mass is the binding energy holding it together. The conversion factor is the speed of light squared. So a mass difference far too small to weigh becomes an energy larger than any chemical reaction can produce. Work in atomic mass units and 1 unit is about 931 MeV. That is what turns a defect of 8 thousandths of a unit per nucleon into more than 1000 MeV for the whole nucleus.

TAKEAWAY 9W

Tiny changes in mass correspond to large nuclear energies.

END OF THE STAGE 15W

Mockups are useful only when their materials, geometry, and diagnostics preserve the question being tested.

Synchronize many channels

PLAN CARD 119W

February 1945, and Kistiakowsky's assembly has failed on the bench twice while every channel checks out alone. Norris Bradbury, who is assembling the test, wants the procedure rewritten before a third attempt; Kistiakowsky wants to know which lot the failures came from. Today you compare the timescales involved, work out what a batch of components actually guarantees, and coordinate many channels as one system. A system of many channels fails in ways no single channel can show, which is why Bradbury's answer — one written procedure, checked the same way every time — catches things no bench test does. What matters is not whether each one works. It is whether they work together, on time, on the same signal.

BRIEFING 28W

A multi-channel system can fail even when every part works separately. Compare the relevant timescales, establish lot-acceptance evidence, and then confront synchronization across the full set of channels.

WHAT YOU DECIDE 13W

Compare physical timescales, qualify initiation hardware, and coordinate many channels as one system.

10.1 Competing timescales

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 40W

Several physical clocks run at once during rapid compression, and the design problem lives in the gaps between them. On a test fixture, two initiation channels have different timestamps, and the team has the measured wave speed and a ruler.

GUIDE 45W

Four numbers, and two of them are channel spreads a hundred apart. Ask of each which spread the measurement gave. And note why this matters at all. Several clocks run at once here, and a timing disagreement becomes a distance the wave has already travelled.

ASK 10W

Estimate how far ahead the early side runs, in millimetres.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a wave crosses a distance in a time set by its speed

VERDICT 84W

Several clocks run at once. Neutron generations multiply on one. The material compresses on another. The assembly begins to blow itself apart on a third. The device works only if multiplication outruns disassembly. These processes differ by many powers of ten, so no single calculation or experiment can hold all of them. The interfaces between them are where it gets hard. This is one: a timing disagreement becomes a geometric error at the speed the wave travels. Half a microsecond is millimetres of asymmetry.

TAKEAWAY 14W

Processes separated by many orders of magnitude often require different numerical or experimental treatments.

10.2 What evidence qualifies a lot?

ORDNANCE & ENGINEERING · A PERSON · VALUE

SCENE 37W

The detonator programme has competing requests and 100 points of test credit. A production lot is waiting for a release decision, while the bench holds one carefully built prototype, inspection records, and access to a full-system trial.

GUIDE 59W

One hundred points of test credit, and a production lot waiting on a release decision. Open each option and ask what it would tell you about the lot rather than about the design. A carefully built prototype and a full-system trial are both evidence about something, and only one kind of evidence is about the units in the box.

ASK 10W

Which measurement tells you whether a production lot is reliable?

BEHIND THE BUTTON 140W

Why a prototype cannot qualify a lot. It was built by the best person available, slowly, with attention nobody can repeat a thousand times. It measures what the design can do, not what the production line did. Those are different questions and the release decision is about the second.

What inspection records can and cannot show. They record that units passed the checks somebody chose. If the failure mode is not among those checks, a perfect record is consistent with a bad lot — which is why sampled functional testing exists at all.

Why the full-system trial is seductive and expensive. It exercises everything at once, so a pass is reassuring and a failure is hard to attribute. Spending most of the credit on one integrated result can leave you knowing that something worked without knowing whether this lot will.

TAKES AS READ

a sample can estimate the spread of a production population · a component test is more diagnostic than an integrated test when the component itself is the question

VERDICT 76W

A single excellent unit can be repeatable and still say nothing about the next units off the line. Lot acceptance is a population claim. The evidence therefore has to sample the population and measure the spread of the quantity that matters. Dimensional inspection checks conformance to a drawing, not timing performance. A full-system test mixes too many causes together. Sampling several lots gives a distribution, exposes tails, and directly answers the decision the programme must make.

TAKEAWAY 20W

Reliability is a property of a manufactured population, so acceptance evidence must measure its distribution rather than its best member.

10.3 Synchronizing many channels

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · BALLPARK

SCENE 41W

The timing-budget meeting has stalled. Each subsystem lead insists that their contribution is small, yet the combined requirement is close enough that nobody wants to sign it. The system cannot move to the next review until one number closes the argument.

GUIDE 47W

Four numbers, and one of them is the other three added straight. Ask whether independent errors add that way. They do not. They combine in quadrature, so the largest term dominates. The straight sum is offered because it is the intuitive answer, and it overstates the total.

ASK 2W

Estimate σ_{total} .

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

independent errors do not simply add

VERDICT 81W

The physical system contributes some delay, cable lengths contribute more, a trigger threshold shifts the moment a pulse is registered, and the recording instrument adds its own. Los Alamos handles this by building a timing budget: every contribution named, measured and combined. Independent contributions add in quadrature — squared, summed, square-rooted — so the largest term dominates and several smaller ones still raise the total. Two, three and four nanoseconds do not make nine; they make about five and a half.

TAKEAWAY 9W

A timing budget must include the full measurement chain.

10.4 Will it hold a chain, or will it not

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM ·
CALLBACK · BELT

SCENE 40W

A shipment has arrived and the vault wants it sorted before anything is stored together. Some of what is on this bench will sustain a chain reaction given enough of it. The rest will not, whatever you do with it.

GUIDE 50W

Two bins. A nuclide that fissions on slow neutrons will hold a chain given enough mass and the right geometry. One that only fissions on fast neutrons, or not at all, will not — however heavy the piece is. Sort on the nuclide, not on how dangerous the label looks.

ASK 15W

Send each item to the bin that says whether it can sustain a chain reaction.

BEHIND THE BUTTON 77W

Why the distinction is the whole of storage policy. Two pieces that each hold no chain can hold one together, so the rule about what may sit next to what is written from this sort. Nothing about mass or radioactivity substitutes for it.

Why radioactive and fissile are different words. Cobalt-60 will make a counter scream across the room and will never sustain anything. Natural uranium is barely active and contains the isotope everything here is about.

TAKES AS READ

some heavy nuclei split when they absorb a neutron · fission, neutron multiplication and chain reactions — taken as read

VERDICT 153W

A chain reaction needs a nuclide that fissions when it absorbs a neutron and then releases enough neutrons to do it again. Uranium-235 and plutonium-239 do that with slow neutrons, so a large enough piece in the right shape sustains itself. Uranium-238 fissions only on fast neutrons and absorbs the slow ones, so it damps a chain rather than carrying one — which is why natural uranium is nine-tenths a moderator by mass. Everything else on this bench is a different property entirely. Cobalt-60 and radium are intensely radioactive and cannot sustain anything. Beryllium and graphite are not radioactive at all and change everything, because they reflect and slow neutrons, which is why they are stored under the same rule. That is the point of sorting this way rather than by how loud a counter gets: the storage policy is written about what can happen when two pieces are put near each other.

TAKEAWAY 16W

What sustains a chain is decided by the nuclide, not by how radioactive a thing is.

END OF THE STAGE 14W

Synchronization is a systems problem involving component variability, signal timing, and shared failure modes.

Build an uncertainty budget

PLAN CARD 97W

March 1945, and this is where Bethe is right and Groves does not want to hear it. More data arrived this month and confidence went down. Today you work out how uncertainties combine, recognise when measurements are not telling you separate things, and assign a number to the whole system. Three of the four measurements in it came off the same calibration. The honest budget names which term dominates and which cannot be reduced by working harder. This is the figure that will be quoted, years later, in a room where nobody remembers what went into it.

BRIEFING 29W

More data do not automatically produce more confidence. Carry uncertain inputs through, identify measurements that share an error, and assign the uncertainty that belongs to the integrated compression claim.

WHAT YOU DECIDE 15W

Combine theoretical uncertainty, measurements that share an error, and system-level evidence into one decision tool.

11.1 Carrying uncertainty through

THEORY & CALCULATIONS · A ROOM · BALLPARK

SCENE 33W

A predicted reaction rate rests on a cross section known to a few per cent and a flux known to a few more. Theory is being asked how well the prediction is known.

GUIDE 46W

Four numbers, and one of them is the two others added straight. Ask whether independent uncertainties add that way. They do not: they combine in quadrature. And note what the answer is for. The larger term dominates, so improving the small one barely moves the total.

ASK 5W

Estimate the total fractional uncertainty.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two independent errors are unlikely to be wrong the same way at once

VERDICT 78W

Independent uncertainties combine in quadrature: square each contribution, add the squares, and take the root. 8% and 6% make 10, not 14. The larger term dominates, so reducing a small contribution may barely change the total while improving the largest one matters much more. That ranking is the practical value of an uncertainty budget. It tells a laboratory with limited people and time which measurement is worth improving first, rather than treating every error bar as equally urgent.

TAKEAWAY 14W

Adding independent uncertainties linearly is usually too pessimistic; ignoring correlation can be too optimistic.

11.2 Where the fragments go

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · CHOICE

SCENE 41W

A nucleus initially at rest splits into a heavy fragment and a light fragment. The team has measured the fragment masses and is arguing about which one should carry the larger share of kinetic energy before looking at the detector data.

GUIDE 50W

Four answers, and the nucleus started at rest. Ask of each whether it respects that. Total momentum was zero, so the two fragments leave with equal and opposite momentum. From there the question is what equal momentum does to two different masses, and one option refuses to decide at all.

ASK 17W

A nucleus at rest splits into a heavy and a light fragment. Which carries more kinetic energy?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

momentum is conserved in an isolated system

VERDICT 72W

The nucleus starts at rest, so total momentum is zero. The two fragments therefore leave with equal and opposite momentum. The lighter fragment must move faster because the momentum magnitude is the same for both. Kinetic energy can be written as $p^2/(2m)$. At equal momentum, the smaller mass has the larger kinetic energy. Momentum conservation fixes the relation between the motions; the energy formula then tells how the released energy is divided.

TAKEAWAY 13W

Momentum conservation fixes the ratio of fragment speeds, and energy follows from it.

11.3 Build the implosion uncertainty budget

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · DIAGNOSIS

SCENE 43W

An implosion trial has returned two anomalies, and the next development cycle can afford only one major push. Division leads are waiting outside the analysis room to learn where the effort goes. Name the wrong source and weeks disappear into the wrong fix.

GUIDE 45W

The panel has more than one thing wrong, so no single cause covers it. Ask of each candidate which readings it leaves unexplained. A budget's value is the ranking rather than the total: improvement starts with the dominant source, and effort anywhere else changes nothing.

ASK 20W

Read the whole reading panel, then select the two causes that together account for every reading. No single cause does.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

two problems can have two different causes

VERDICT 88W

A budget lists the separate contributions: material properties, geometry, timing, diagnostics, models. For each one it estimates how much it moves the final claim. Its value is not the total but the ranking. Improvement starts with the dominant source, and effort spent anywhere else changes nothing. The honesty of it matters here too. When two anomalies sit in different subsystems, the budget has two entries. Force

one cause onto both and you misattribute one of them — and that is the entry somebody then goes and works on.

TAKEAWAY 9W

Improvement starts by naming the dominant source of uncertainty.

11.4 Radiochemical tracers

CHEMISTRY & METALLURGY · A ROOM · CALLBACK · BALLPARK

SCENE 30W

Chemistry and Metallurgy is running separations on quantities too small to weigh and needs to know how much is being lost. Every percent lost is weeks of production at Hanford.

GUIDE 50W

Four numbers, in two pairs, and only one pair belongs to this separation. Ask of each whether it is what went in or what came out. And note what makes a raw ratio dishonest here. The tracer decayed throughout, so both counts have to be corrected to the same moment.

ASK 4W

Estimate the chemical yield.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a radioactive sample announces itself through its own radiation

VERDICT 81W

A tracer is a small amount of radioactive material that travels with the chemistry and reports where it went. That turns an invisible loss into a measurable fraction: yield is what came out over what went in. The correction that makes it honest is decay — the tracer has been decaying throughout the separation, so a raw count afterwards understates the recovery and flatters the loss. Both counts have to be corrected to the same moment before the ratio means anything.

TAKEAWAY 10W

A tracer turns invisible material loss into a measurable fraction.

END OF THE STAGE 16W

Uncertainties must be traced through the full evidence chain, especially when measurements share hardware or assumptions.

Design the integrated non-nuclear campaign

BEFORE THE STAGE — EVADE

The security officer with questions about your notebook

A counter-intelligence officer has been waiting outside the laboratory since eight about a notebook you took to the wrong building, and everybody here has learned that the conversation is shorter with a division head present. Stay out of his way until Oppenheimer, the laboratory director, is back.

PLAN CARD 95W

April 1945. The whole schedule now runs through one machine shop and nobody had drawn that until Bacher's committee sat down to force the interfaces. Groves wants a date; Bacher will not schedule a test he cannot instrument. Today you set what counts as a passing test, instrument it so the result can be interpreted, and find what the schedule really depends on. A test earns its place by telling two explanations apart, not by being impressive to watch. Bacher's rule for this laboratory is the shortest one on the mesa: no test without diagnostics.

BRIEFING 31W

The project is ready to combine subsystems without nuclear material. Define what counts as acceptable, preserve interpretable measurements, schedule the dependencies, and make the campaign answer the most important unresolved question.

WHAT YOU DECIDE 17W

Set acceptance criteria, instrument the trial, identify the critical path, and choose tests that discriminate among explanations.

12.1 Acceptance criteria

CHEMISTRY & METALLURGY · A ROOM · BALLPARK

SCENE 29W

A property has to be at least 100 units and the measurement reads 102, plus or minus 3. Material this scarce cannot be scrapped casually, or accepted on hope.

GUIDE 50W

Four numbers, and two of them are uncertainties ten times apart. Ask of each which one this measurement carries. And note what gets compared with the threshold. Not the central value, but the lowest the result could plausibly be. That rule has to be agreed before the number is known.

ASK 8W

Estimate the conservative lower bound of the measurement.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a measurement has a range, not just a value

VERDICT 77W

An acceptance criterion turns a desired property into a rule: a limit, a measurement method, and a stated uncertainty. Results near the boundary are why the uncertainty rule must be agreed before the number is known. Here the central value is 102, but the lower plausible value is 99. That falls below the 100-unit requirement, so the measurement does not demonstrate a clean pass. Reading only the midpoint treats an uncertain measurement as though it were exact.

TAKEAWAY 10W

Acceptance criteria need explicit rules for uncertainty and borderline results.

12.2 Buy the test that can decide

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · VALUE

SCENE 41W

Three team leads are waiting outside the facility office, each with a test they say cannot wait. There is one week of access left in the schedule, and only one proposal can take it. The scheduler needs one name before Monday.

GUIDE 52W

One week of access, three tests, and one name needed before Monday. Open each proposal and ask what decision its result would change. A test that everybody agrees is important but whose result changes nothing this month has not earned the week. Commit to the one that moves the programme's next decision.

ASK 7W

Which test changes the programme's next decision?

BEHIND THE BUTTON 108W

What makes a test decisive rather than valuable. Two possible outcomes leading to different actions. A test whose likely result confirms the current plan is worth running eventually and not worth the only week of access left.

Why the leads all say theirs cannot wait. Each is right about their own programme and none of them is holding the schedule. The scheduler is the only person looking at what the next decision actually requires, which is why the choice sits there.

Why Monday. A name given late costs the week itself, so an imperfect choice made in time beats a better one made after the access has lapsed.

TAKES AS READ

an experiment is useful when different explanations predict different outcomes · facility time is limited

VERDICT 75W

Before a test is run, each hypothesis should predict what the chosen observable will do. If both hypotheses predict the same trace, even perfect data cannot separate them. Repeating one condition can reduce statistical uncertainty while leaving the decision unchanged. A large integrated shot can also be ambiguous if several causes move the same output. The valuable test is the smallest one whose outcomes differ between the hypotheses, with diagnostics aimed directly at that difference.

TAKEAWAY 18W

A test is valuable when its possible outcomes distinguish competing explanations, not when it merely produces more data.

12.3 Critical path to an integrated trial

ORDNANCE & ENGINEERING · A ROOM · BALLPARK

SCENE 39W

Three chains of work are running in parallel, and the integrated trial cannot happen until the longest one finishes. Fat Man's schedule is managed this way: the calendar is controlled by dependencies, not by how many tasks look busy.

GUIDE 52W

Four numbers, and two of them belong to other questions: one task, and both chains added together. Ask of each whether it is a chain length. Only the longest chain moves the date. The gap between the two is what a parallel chain can lose before it becomes the one that matters.

ASK 13W

Estimate how much a parallel chain can slip before it controls the date.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

work that can happen in parallel does not add up

VERDICT 88W

The longest chain is the critical path. It is the only place where adding people or shifts moves the date. Effort poured into a task that is not on it changes nothing, however busy it makes everyone. It is spent on float that was already there. Float is the gap between the controlling chain and the one being worked on, and it is finite. Enough slippage on a parallel chain makes that chain the new critical path. At which point the whole calculation has to be done again.

TAKEAWAY 8W

Schedule management starts with dependencies, not activity counts.

12.4 What a scaled shot can and cannot tell you

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · CHOICE

SCENE 39W

A half-scale shot is proposed: an eighth of the explosive and a much shorter schedule. The material detonation velocity is listed at both scales, and the team has geometric dimensions, reaction-zone measurements, and timing records spread across the table.

GUIDE 53W

Four claims about a half-scale shot. Ask of each what it assumes scales. Ratios do, and the phenomena they govern come with them. A detonation velocity, a diffusion length and a reaction time do not, because halving the geometry does not halve them. So one option is too generous and one too harsh.

ASK 13W

A half-scale shot is fired. Which of its results transfer to full scale?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a dimensionless ratio has the same value at any scale · units and conversions: MeV, u, barns, scientific notation — taken as read

VERDICT 81W

Scaling works through dimensionless groups: reproduce those and the phenomena they govern reproduce too. What does not scale is anything with a dimension of its own — a detonation velocity, a diffusion length, a reaction time — because halving the geometry does not halve them. So a half-scale shot can settle a symmetry question governed by ratios and cannot settle a timing margin measured in

microseconds. The discipline is to state, before firing, which quantities the shot is entitled to decide.

TAKEAWAY 14W

A scaled experiment reproduces the dimensionless quantities, and only what depends on them transfers.

12.5 What the shop is making today

THEORY & CALCULATIONS · A ROOM · SPOT

SCENE 47W

The machine shop takes work from every division and cannot take it all. The priority is set on a card at the window, and it is changed when the schedule changes — after the test date moves, after the assay comes back, after somebody senior walks in.

GUIDE 51W

Release the jobs the current priority wants and hold the rest. The card at the window is changed during the day and nobody announces it. What is scored is the jobs either side of a change, because they are the only ones that show whether the window is reading the card.

ASK 12W

Release to the priority on the card, and keep watching the card.

BEHIND THE BUTTON 100W

Why a shop priority changes. Early it is whatever the test date needs. When an assay comes back wrong it is whatever the re-measurement needs, because a number nobody trusts stops every calculation resting on it. When a division head appears it is whatever that division wants, which is a different rule and worth naming honestly.

Why the changeover costs here. A machined part is hours of a scarce machinist and a scarcer piece of material. A job released under the old priority is not merely late — it has consumed the capacity the new priority was written to buy.

TAKES AS READ

a workshop with limited capacity decides what to make first

VERDICT 131W

Three priorities run across the day and each wants a different part of the queue. Anything the test date needs, then anything a re-measurement needs, then anything a division head has asked for in person. A job can answer two at once, which is what makes the change cost real. The panel scores the window either side of each change, because most of the queue is wanted by neither priority and is correctly held by somebody who has read nothing. And the cost here is material as well as time: a part released under yesterday's priority has used a machinist, a lathe and a piece of metal that the new priority was written to spend on something else, and none of the three comes back at the end of the day.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE STAGE 14W

An integrated trial succeeds when criteria, diagnostics, schedule, and scientific discrimination are designed together.

Treat Trinity as an experiment

PLAN CARD 91W

July 1945, and there is 1 shot with weather closing in. Groves wants the date held; Bethe wants the questions written down before anybody drives to the tower — and this time it is Groves who has to give way, because a test whose questions are decided afterwards answers none of them. Today you state those questions in advance. You build the field arrangements around the measurements, rather than the other way round. And you cover the ways the instruments themselves can fail. There is no second attempt to design around.

BRIEFING 35W

A dramatic event is not automatically a useful experiment. Specify the questions in advance, build the field system around the evidence needs, cover the failure modes with diagnostics, and establish how predictions will be judged.

WHAT YOU DECIDE 18W

Define what the integrated test must answer, align field engineering with diagnostics, and validate predictions against independent evidence.

13.1 What Trinity had to answer

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · PROTOCOL

SCENE 45W

Trinity is being written as a set of questions before it becomes an event. One full-system attempt will carry dozens of expectations, and every division wants its own result treated as decisive. There will be no second shot to clarify what the first one meant.

GUIDE 47W

Four questions and four kinds of evidence. Pair them by asking what each question needs: the whole event, one subsystem's traces, a comparison with something written beforehand, or measurements taken afterwards. A test without its questions written down first becomes an argument later about what it proved.

ASK 15W

Match each question Trinity must answer to the evidence that can actually settle that question.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a test proves what it was designed to ask

VERDICT 81W

A test without its questions written down in advance becomes an argument afterwards about what it proved. Different questions need different evidence. Full-system nuclear performance requires the integrated event. Firing-system coordination comes from timing records and can be judged even if another part disappoints. Comparing observed effects with pre-test predictions tests the model. Safety evidence addresses hazards at and beyond the site. Writing those questions

first keeps one dramatic outcome from swallowing every other claim the experiment was designed to answer.

TAKEAWAY 16W

A full-system test should be written as a set of questions before it becomes an event.

13.2 Trinity field engineering

ORDNANCE & ENGINEERING · A PERSON · SEQUENCE

SCENE 46W

The desert crew has the tower, cables, cameras, power, and communications spread across a schedule that is running out of room. Bainbridge has a forecast that closes the window in four days, whatever is unfinished. Every team wants to know whether the field is truly ready.

GUIDE 45W

All four will happen, and field work succeeds or fails on interfaces. Ask of each card what has to be proved before the thing that depends on it arrives. An interface inspected while replacement is still possible is a different thing from one inspected afterwards.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a schedule set by weather cannot be extended

VERDICT 90W

Field integration succeeds or fails on interfaces, and each interface has to be proved before the thing that depends on it arrives. Site survey, power, communications, and controlled access come first. Electrical and diagnostic interfaces are installed and independently inspected while replacement is still possible. Only then does the approved assembly enter the field configuration. The countdown runs against defined hold points, and the site is secured and documented afterwards. The order preserves the ability to find and correct a failed interface before it becomes buried inside the next step.

TAKEAWAY 17W

Field integration is ordered so that every interface is proved before the thing that needs it arrives.

13.3 The plot that makes a decay a straight line

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · CHOICE

SCENE 30W

Two counters have produced a table of count rate against time, and the notebook has it plotted on ordinary axes as a curve that nobody can read a number off.

GUIDE 57W

Four ways to plot the same data. Ask of each what shape it turns an exponential into. Then ask what the slope of that shape means. One option takes the logarithm of both axes, which suits a power law rather than a decay. And a straight line is easy to check by eye, which is the point.

ASK 14W

Which transformation makes exponential decay linear, and what physical quantity comes from the slope?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a logarithm turns a constant ratio into a constant difference

VERDICT 76W

An exponential plotted on linear axes is a curve, which is hard to check by eye. Take the logarithm of the count rate and the same data becomes a straight line. Its slope is minus the decay constant, so the half-life follows from $t^{1/2} = \ln 2 / \lambda$. The transformed plot also exposes failures: a second component can change the slope, while background can flatten the line at late times. Choosing the axes is part of the measurement.

TAKEAWAY 17W

Plotting a logarithm turns an exponential into a straight line, and its slope is the decay constant.

13.4 Where the energy in fission comes from

THEORY & CALCULATIONS · A ROOM · CHOICE

SCENE 32W

A visitor asks why splitting a heavy nucleus releases energy while splitting a light one costs energy. The blackboard has the binding-energy curve on it and nobody has used it to answer.

GUIDE 52W

Four accounts of where the energy comes from. Ask of each whether it uses the shape of the binding-energy curve. It rises steeply, peaks near iron, and falls slowly to uranium. So what matters is whether the products sit higher on it than the reactants, and two of these options ignore that.

ASK 12W

Why does splitting uranium release energy while splitting carbon would absorb it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a nucleus is held together by binding energy

VERDICT 87W

Binding energy per nucleon rises steeply from hydrogen, peaks around iron and nickel, and falls slowly to uranium. A reaction releases energy when its products sit HIGHER on that curve than its reactants, because the products are more tightly bound. Uranium splitting into two mid-mass fragments moves up the curve, so energy comes out. Carbon splitting would move down it, so energy would have to go in. The same curve is why fusion of light nuclei also releases energy — from the other side of the peak.

TAKEAWAY 11W

The binding-energy-per-nucleon curve says which way a nuclear reaction releases energy.

13.5 Close the material balance

CHEMISTRY & METALLURGY · A ROOM · CALLBACK · BALANCE

SCENE 38W

A separation batch has come back and the numbers do not close. Downstream, somebody is going to cast this material assuming the report meant what they thought it meant, so Chemistry has to close the ledger before release.

GUIDE 49W

Read every stream first, because reading costs nothing. Then count only the masses that belong to the same material ledger. The batch does not close, and the missing mass has gone somewhere real — an uncounted route rather than an arithmetic slip. Report the total when the ledger closes.

ASK 7W

Which uncounted route closes the material balance?

BEHIND THE BUTTON 130W

Why a balance has to close. Material is conserved, so a batch that does not add up is a statement that something has not been measured, not that some of it stopped existing. Every unclosed ledger is a location nobody has looked at yet.

Where material actually goes. Holdup in the equipment, residue on filters and vessel walls, a side stream taken for analysis, losses to a scrubber. Each is ordinary and each is invisible in the streams somebody thought to weigh.

Why it matters downstream. Somebody is going to cast this material assuming the report means what they think it means. In this programme, an assay that is quietly wrong about how much is present is the kind of error that ends with a criticality rather than a memo.

TAKES AS READ

percent of input material can be added across mutually exclusive output routes · purity and recovered quantity are different measurements

VERDICT 69W

The input is 100% of the target material. Recorded product and aqueous output account for only 93.9%, so 6.1% is still missing. The 91% purity result answers a different question and cannot be added to that ledger. A wall swab finds 5.8% of the input on the vessel surface. Add that loss route and the balance closes to 99.7%, inside rounding. Yield, purity, and material balance are different claims.

TAKEAWAY 19W

A purity result cannot substitute for a closed material balance; losses outside measured process streams must still be counted.

A decisive test requires predeclared questions, independent diagnostics, and a comparison between predicted and observed outcomes.

Reconcile evidence and freeze the design

PLAN CARD 133W

Late July 1945. Enrico Fermi dropped shredded paper from the bunker and got about ten kilotons from how far the blast wave carried it. That sits outside the band the theory division predicted, and either the model is wrong or the measurement is — with Bethe, who has already corrected several overestimates on paper, on one side of it. Today you compare prediction against observation properly, close the materials record, and decide what may still change. The materials record has to close as carefully as the physics, because a number is only about a thing if you know which thing it was measured on. A design freeze is an evidence decision that happens to have a date on it, and Oppenheimer is the one who has to hold the room to that distinction.

BRIEFING 38W

The test has produced data, but the program still must decide what those data mean. Compare observations with uncertainty bands, verify the material record, resolve disagreements between model and measurement, and freeze only the changes supported by evidence.

WHAT YOU DECIDE 15W

Compare prediction bands with observations, release materials, reconcile theory and experiment, and control post-test changes.

14.1 Compare a prediction band with an observation

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · BALLPARK

SCENE 30W

A model predicts an observable between 40 units. The measurement reads 21, plus or minus 2. The temptation after Trinity is to read one comparison as a verdict on everything.

GUIDE 48W

Four numbers, and two of them describe the band rather than the measurement: its midpoint and its lower edge. Ask of each what the comparison needs. A gap is measured in units of the measurement's own uncertainty. And note that a wide band would have accommodated almost anything.

ASK 15W

Estimate the offset in units of the measurement uncertainty, then state the strongest justified conclusion.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a wide prediction is easy to agree with

VERDICT 83W

The gap between an observation and a prediction is measured in units of the measurement's own uncertainty. Take the difference and divide by that uncertainty. Half a sigma from the middle of the band is consistent. But consistency with a band this wide is weak evidence, because 15 to 25 would have accommodated almost anything. Agreement supports the part of the model that produced the band, and no more. One agreeing diagnostic can sit beside others that disagree, and those still need explaining.

TAKEAWAY 13W

Evidence should update a model claim by claim, not by declaring total victory.

14.2 Why separation is a physical problem

CHEMISTRY & METALLURGY · A PERSON · CHOICE

SCENE 41W

At Chemistry and Metallurgy, a production planner brings a request to enrich uranium-235 relative to uranium-238 by an ordinary chemical route. The division has been asked to decide whether the two isotopes give chemistry enough contrast to make that plan practical.

GUIDE 51W

Four explanations, and all four mention a difference. Ask of each whether that difference is chemical or physical. The electron structures are almost identical, which is what ordinary chemistry answers to. What is left is about one part in 235 of mass, and that is what a physical process can grip.

ASK 14W

Why is ordinary chemistry a poor way to separate uranium-235 from uranium-238 in bulk?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

chemical behaviour is set by electrons, and isotopes differ only in the nucleus

VERDICT 77W

Chemical behaviour is controlled mainly by electrons, and the two uranium isotopes have the same electron count and almost the same electronic structure. Their ordinary chemical reactions are therefore nearly identical. The useful contrast is the small mass difference, about 1 part in 235, which can be exploited by processes that respond to mass or mass-to-charge ratio. The tiny size of that contrast is why isotope enrichment requires repeated separation stages rather than one ordinary chemical reaction.

TAKEAWAY 20W

Isotopes of one element have nearly the same electron structure, so practical enrichment must exploit a tiny isotope-dependent physical difference.

14.3 Spend the follow-up checks on independence

THEORY & CALCULATIONS · A ROOM · VALUE

SCENE 33W

Theory and experiment disagree by three standard deviations. The division can complete two follow-up checks before the next review, and every extra run that repeats the same dependency consumes one of those chances.

GUIDE 50W

Two follow-up checks, and a three-sigma disagreement between theory and experiment. Open each check and ask what it depends on. A run that repeats a dependency the disagreement already contains cannot resolve it, however much data it returns. Buy the two that could change the interpretation rather than the precision.

ASK 10W

Which two checks can change the interpretation of the disagreement?

BEHIND THE BUTTON 129W

What three standard deviations is and is not. It is unlikely to be chance and it is not proof of new physics. Most disagreements that size turn out to be a shared assumption — a cross-section, a calibration, a background model — used on both sides or wrong on one.

Why repeating the measurement rarely helps. More runs on the same apparatus reduce the statistical part of the error and leave every systematic exactly where it was. If the disagreement is systematic, better statistics make it sharper and no more explicable.

What independence looks like here. A check that uses a different detector, a different reaction, or a different calibration path constrains the same quantity through different assumptions. Only that can tell a real discrepancy from a shared mistake.

TAKES AS READ

a systematic error does not average away with more repeats · independent measurements should not share the same critical calibration

VERDICT 77W

More of the same data reduces random error but carries the same systematic dependencies. Retuning the model can hide the discrepancy without explaining it. First audit the calibration shared by the measurements, because a common bias can create a confident disagreement. Then use a measurement based on different hardware or a different calibration path. If the discrepancy survives independent evidence, the case for missing physics strengthens. If it disappears, the original measurement chain becomes the leading suspect.

TAKEAWAY 19W

When theory and experiment disagree, the next measurements should attack shared systematics and add evidence based on different dependencies.

14.4 Post-Trinity design freeze

ORDNANCE & ENGINEERING · A ROOM · SEQUENCE

SCENE 43W

Trinity has produced new evidence, and a last-minute improvement is already being argued over. Drawings are moving toward production while engineers are still proposing changes. The room has one uncomfortable question: what can still change without severing the link to what was tested?

GUIDE 52W

All four will happen, so ask what each one is the input to. There is nothing to compare with a prediction until the separate records agree with each other. And nothing can be frozen before the evidence has been weighed. Naming what stays unexplained is part of the comparison, not an afterthought.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

what was tested and what gets built can drift apart

VERDICT 84W

Each stage is the input to the next. The timing, hydrodynamic and radiation records are reconciled into one account of the shot, because there is nothing to compare until they agree with each other. That account is then compared with the pre-test predictions, and what remains unexplained is named rather than absorbed. Only then can the evidence-supported changes be approved and the configuration frozen. The freeze comes before the paperwork — reverse those two and the documents describe a configuration that is still moving.

TAKEAWAY 13W

A freeze comes before the paperwork, or the paperwork describes something still changing.

14.5 Find what carries the background

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A ROOM · CALLBACK · CONTROL

SCENE 36W

A weak sample has been counting for an hour and the question is whether it is emitting anything at all. This is where an experimental group either establishes a real signal or convinces itself of one.

GUIDE 75W

The number you are watching is the counting rate with no sample in the station, and the excess is how much of it cannot be accounted for. A change has told you something only when it moves that excess by more than the rate wanders on its own between runs. Several parts of the station changed when it was built, so choose one, run it, and put it back. Name the part the excess follows.

ASK 9W

Which part of the counting station carries the excess?

BEHIND THE BUTTON 178W

What a background is. A detector counts things whether or not you gave it anything to count: cosmic rays, activity in the concrete, contamination on the housing, noise in the electronics. The rate with no sample in place is the sum of all of those, and a weak sample is only detectable to the extent that its own counts stand above that sum.

Why one change at a time. Change two things and a response tells you only that one of them mattered. The station has a shield, a room and a detector, and each of them can carry background for its own reason, so the only reading that names a part is the one where nothing else moved.

Why putting it back matters. A rate that drops when you shield the detector and stays down when you unshield it was falling for some other reason — drift, a decaying source, somebody switching off a pump down the corridor. A cause has to follow the change in both directions, and that is the whole of what reversal tests.

TAKES AS READ

background is measured with no sample present · one changed variable can be tested against a stable baseline

VERDICT 78W

The sample cannot explain a count excess that remains after the sample is removed. Extra shielding cannot explain it when more lead changes nothing. The decisive comparison is hardware: the original detector still runs high in a clean room, while an identical clean detector in the same room returns to the normal background. Put the original detector back and the excess returns. That reversal makes contamination of the detector or holder a causal result rather than a correlation.

TAKEAWAY 19W

A suspected cause is strongest when changing only that cause moves the reading and reversing the change restores it.

END OF THE STAGE 11W

Design freeze is an evidence decision, not merely a calendar date.

Complete the evidence chain—and confront responsibility

BEFORE THE STAGE — TAG

Catch Bethe before he goes into the Tuesday colloquium

Bethe, who heads the Theoretical Division, has the corrected cross-section figure, and once the colloquium starts he is at the board for two hours. Every estimate on your desk this afternoon is built out of that number, and the version you have is the one from before he found the error.

PLAN CARD 114W

August 1945. The technical questions are closing and the other ones are not. Today you decide what each line of work is entitled to claim — what it knows, what is still uncertain, how strongly each claim is held — and then the same people have to say what follows from having done the work at all. The second part has no method. Leo Szilard, who wrote the letter that started the programme, circulated a petition in July with 155 signatures on it. Oppenheimer spent the rest of his life being asked. The scientists who were here disagreed about this in public for decades, and never about the physics, which they all agreed on.

BRIEFING 36W

The final mission asks each scientific line to state what it knows, what remains uncertain, and what follows from the work. Complete the theory, diagnostic, and materials dossiers, then address the responsibilities scientists debated after Trinity.

WHAT YOU DECIDE 17W

Close the technical reviews while recognizing that scientific completion does not settle the human and political consequences.

15.1 Why light nuclei release energy too

THEORY & CALCULATIONS · A ROOM · CHOICE

SCENE 40W

The binding-energy curve on the board rises steeply from light nuclei toward iron. A younger theorist has noticed that fusion moves uphill on that same curve and asks why an energy-releasing reaction does not simply happen whenever light nuclei meet.

GUIDE 50W

Four accounts of why fusion is hard. Ask of each whether it is about the energy released or about getting there. Two positive nuclei repel, and the force that binds them reaches only about a femtometre. So the difficulty is closing that gap, which costs temperature rather than a neutron.

ASK 10W

If fusion releases energy, why is it difficult to start?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

like charges repel, and the nuclear force acts only at very short range

VERDICT 84W

Fusion moves up the binding-energy curve from the light end, so it releases energy — more per nucleon than fission does. The difficulty is getting there: two positive nuclei repel, and the nuclear force that binds them only reaches across about a femtometre. Closing that gap needs kinetic energy, which means temperatures of millions of kelvin, whereas fission needs only a slow neutron and no barrier at all. Energy released and ease of ignition are separate questions, and confusing them is the usual error.

TAKEAWAY 21W

Fusion of light nuclei releases energy for the same reason fission of heavy ones does, and it is harder to start.

15.2 Final diagnostic readiness review

EXPERIMENTAL PHYSICS & DIAGNOSTICS · EXPERIMENTAL PHYSICS · A PERSON · SEQUENCE

SCENE 41W

The final readiness review asks whether the instruments will actually record what happens. There is exactly one event to record, no repeat shot to recover a missed observable, and a diagnostic that saturates or loses timing can erase an answer permanently.

GUIDE 49W

All four steps happen, so ask what each one proves that the others cannot. An instrument nobody can describe afterwards produces data nobody can use. A chain of working parts is not a working chain. And independent coverage means one failure does not take a whole answer with it.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

an instrument can be working and still miss the event

VERDICT 83W

Calibration and configuration records come first. An instrument nobody can describe afterwards produces data nobody can use. End-to-end timing and health checks then prove the chain works as a chain, rather than as a set of working parts. Independent coverage of the critical observables comes next — independent, so one failure does not take a whole answer with it. Deployment is approved with the residual risks named and a backup for each. Readiness means failures will be recognised, not that there are none.

TAKEAWAY 14W

The final product is a trustworthy evidence system, not merely a collection of sensors.

15.3 Leave a materials capability, not just better numbers

CHEMISTRY & METALLURGY · A ROOM · VALUE

SCENE 33W

The final dossier has to show more than that components existed. The division has one last hundred credits of improvement to spend, and the choice says what it thinks a materials capability is.

GUIDE 51W

One hundred credits, and the choice says what the division thinks a materials capability is. Open each investment and ask what it leaves behind when the programme ends. Better numbers on the current components are worth something; a method, a standard or a trained group is worth it next year too.

ASK 8W

Which investment creates the most durable materials capability?

BEHIND THE BUTTON 120W

What a capability is, as distinct from a result. A result answers today's question. A capability answers questions nobody has asked yet: a calibrated technique, a documented procedure, a set of reference materials, people who know how to use them. The dossier is being read by people deciding what to fund next.

Why the temptation is to buy numbers. Precision on an existing measurement is visible, defensible and easy to write up. It is also spent the moment the component it describes is superseded.

Why this is the last hundred credits. Nothing follows it, so anything that needs a second round to be useful is worth nothing here. That constraint is what separates the durable investments from the promising ones.

TAKES AS READ

future batches may differ from today's batch · a record is useful when it preserves relationships, not only isolated measurements

VERDICT 78W

Another measurement can describe one batch more precisely, but it does not explain why the next batch differs. A process-to-properties record connects what was done to what the material became. That relationship can be tested across batches and used later to diagnose a change in structure or performance. Extra composition checks, tighter remeasurement, and higher throughput all have value, but none creates that causal record. The durable capability is the evidence chain linking process history to measured properties.

TAKEAWAY 15W

A durable materials programme links process conditions to composition, structure, properties, and traceable batch records.

15.4 Scientific responsibility after Trinity

IMPLOSION & SYSTEMS INTEGRATION · IMPLOSION & INTEGRATION · A ROOM · CHOICE

SCENE 40W

After Trinity, reports of yield, fallout, and measurement uncertainty are spread across the laboratory. Officials outside the site are asking what the results mean. Scientists now have to decide what belongs in their technical assessment before it leaves Los Alamos.

GUIDE 50W

Four ways to draw one line. Ask of each what it gives away. One keeps the laboratory to raw readings, which leaves officials judging consequences nobody has shown them. One treats knowing the facts as authority to decide. The useful version separates the account from the recommendation and labels both.

ASK 14W

Which statement best separates the laboratory's technical responsibility from the political decision that follows?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a technical question and a decision about its use are different questions

VERDICT 79W

Scientists can report what the instruments measured. They can estimate likely effects and state how uncertain those estimates are. That information matters because officials cannot judge consequences they have never been shown. Scientists can also explain which conclusions are well supported and which depend on assumptions. They do not get the final policy decision simply because they know the

technical details. Their responsibility is to make the evidence and uncertainty clear before others decide what to do with it.

TAKEAWAY 17W

Technical expertise carries a duty to communicate consequences and uncertainty clearly without turning expertise into political authority.

15.5 Why Thin Man was abandoned

ORDNANCE & ENGINEERING · A ROOM · CALLBACK · SEQUENCE

SCENE 34W

Thin Man was the plutonium gun design and much of the site's effort was pointed at it. Then the counters returned a number the concept could not survive, and within weeks the laboratory reorganised.

GUIDE 50W

These four are one chain, from a measurement to a decision that cost a year. Ask of each card what has to be known before it. Nothing is unacceptable before it is understood, and nothing is understood before it is measured. The reorganisation is the consequence rather than the argument.

ASK 12W

Arrange the four cards from earliest cause or prerequisite to latest result.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

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TAKES AS READ

a measurement can rule out a design without anybody being wrong · pressure, temperature and equations of state — taken as read

VERDICT 81W

This is what a measurement invalidating a development path looks like from inside: not a debate about opinion, but a chain from one new fact to a decision that costs a year. The unexpectedly high spontaneous-neutron background in reactor plutonium comes first, because it is the new evidence. That makes a slow assembly vulnerable to starting early. That vulnerability makes the plutonium gun unacceptable. And only then does the effort move to implosion, which is a decision rather than a fact.

TAKEAWAY 12W

In complex engineering, one new measurement can invalidate an entire development path.

Technical readiness and historical responsibility are different judgments; a complete curriculum must examine both.

THE CLOSING CARD

The technical questions closed in August 1945, and every line of work could finally say what it knew and how well it knew it. The war ended. The mesa emptied over the following year: the theorists went back to universities, the metallurgists to industry, and the site to a laboratory that is still there.

The physics did not stay secret, because physics does not. What the people here could defend, claim by claim, is written down — and so is what they argued about once they could argue in public. That record is the part you inherit.

Card text only — no options, boards or answers. 12,771 words of card prose across 64 stops. Generated from `themes/project` by `tools/make-cardbook.mjs`.